
AI - POWERED DIGITAL TWIN FOR POWER GRIDS

NEUROGRID

DETECT · PREDICT · DECIDE · ACT

A self-healing, AI-driven digital twin of power grid infrastructure
— simulating real-time operations, predicting failures before
they occur, and autonomously rerouting power to prevent
outages.

HUMMINGBIRDS

THE PROBLEM.



REACTIVE OPERATIONS

Power grids operate in a reactive model

→ Failures are addressed only after outages occur

NO PREDICTIVE INTELLIGENCE

Unable to detect early signs of failure

→ Transformer and line failures go undetected until breakdown

MANUAL FAULT HANDLING

Human intervention required at every step

→ Slower response time and higher risk of human error

ECONOMIC LOSSES

High downtime impacts industries & daily life

→ Unplanned outages translate to massive financial and social costs

FRAGMENTED MONITORING

No unified real-time view of grid health

→ Siloed systems create blind spots in critical infrastructure

DYNAMIC LOAD CHALLENGES

Difficulty managing renewables & spikes

→ Sudden fluctuations increase failure risk across the network

THE INNOVATION JOURNEY.



Grid Digital Twin

Simulates and visualizes real-time power grid performance.



ML-Powered Predictive Intel

Anticipates potential anomalies and service disruptions.



Cognitive Decision Core

Intelligently evaluates options and recommends optimal response.



Adaptive System Recovery

Executes automated, self-healing rerouting for resilience.



Real-Time Operations Dashboard

Provides comprehensive grid health visibility and closed-loop control.

TRANSFORMATION: THEN & NOW

CURRENT SYSTEM

REACTIVE

MANUAL

DELAYED

NEUROGRID

PREDICTIVE

AUTOMATED

SELF-HEALING

CORE INNOVATION

ENGINEERING PARADIGM



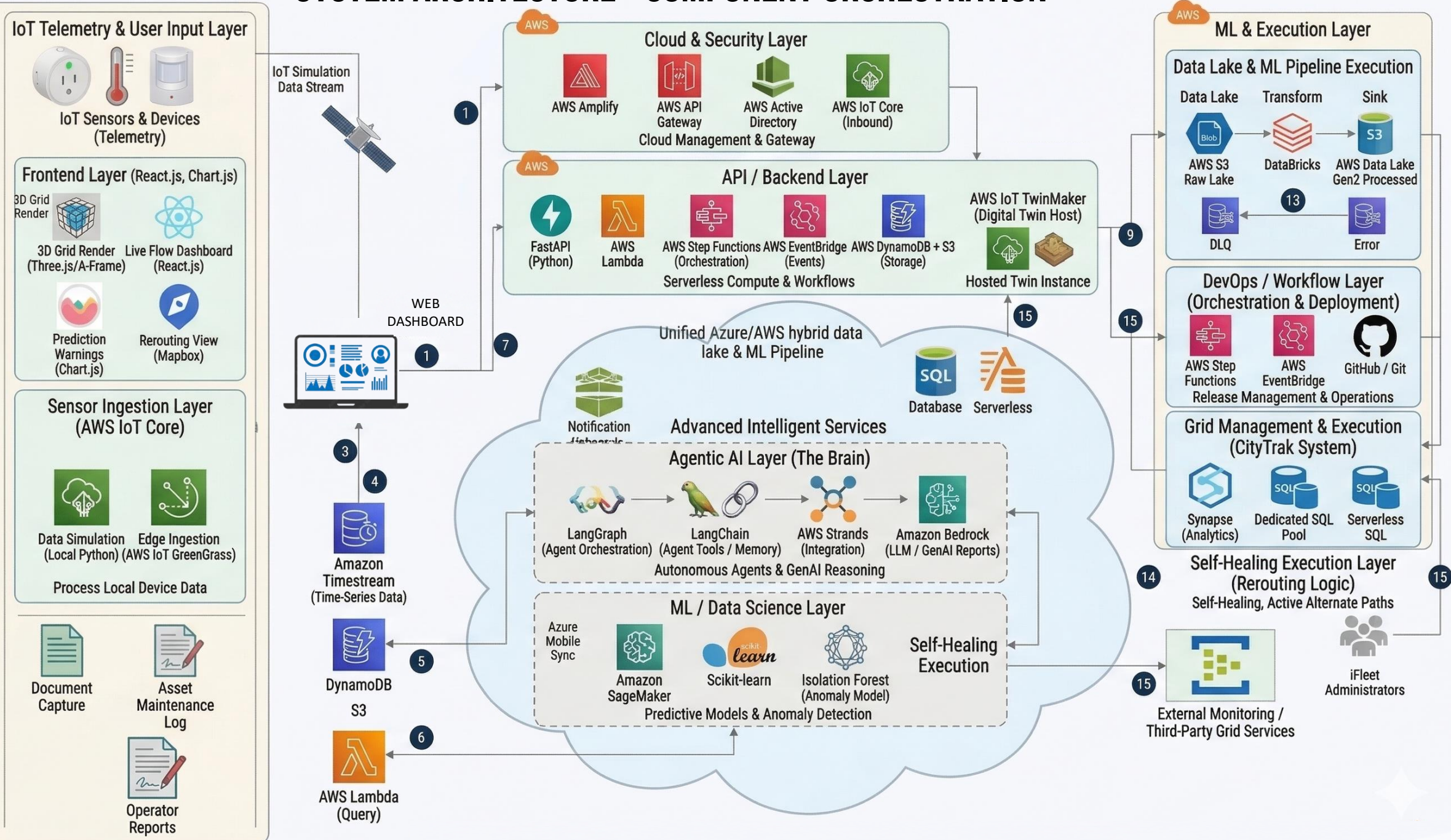
Unlike traditional fragmented systems, the closed-loop engineering closes the circle: from continuous monitoring to automated, proactive grid stabilization.



SYSTEM ARCHITECTURE – COMPONENT ORCHESTRATION

LOCAL / SIMULATION ENVIRONMENT

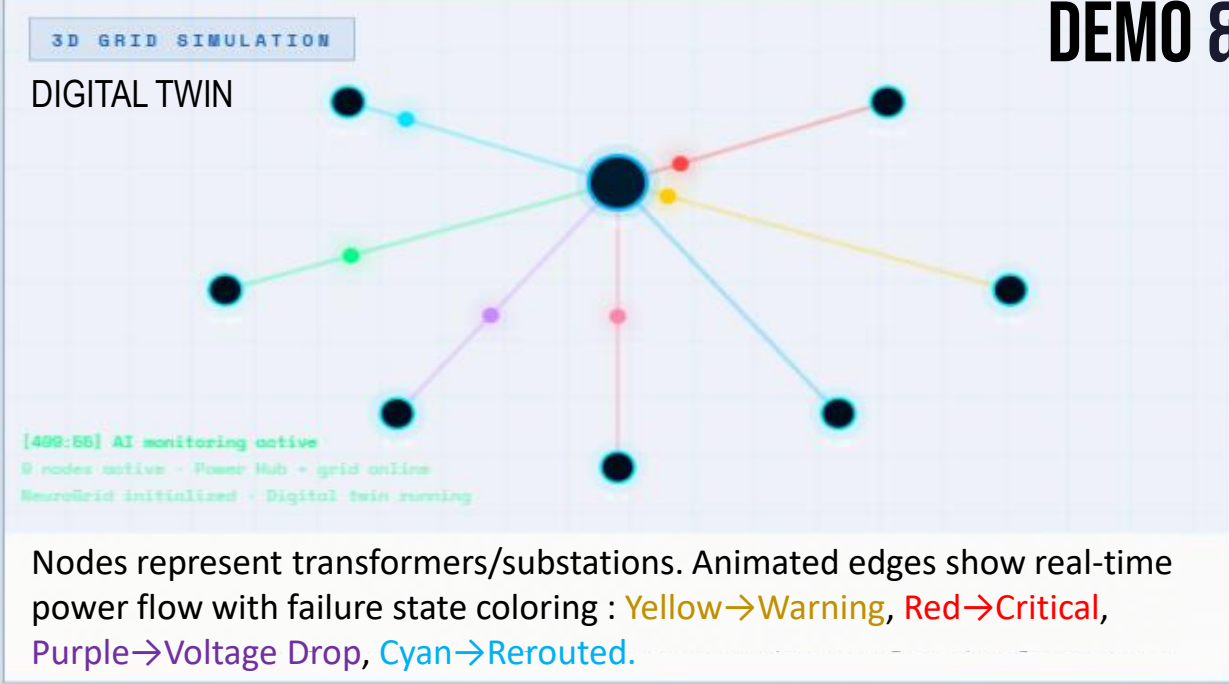
DIGITAL TWIN – TO SIMULATE TELEMETRY DATA



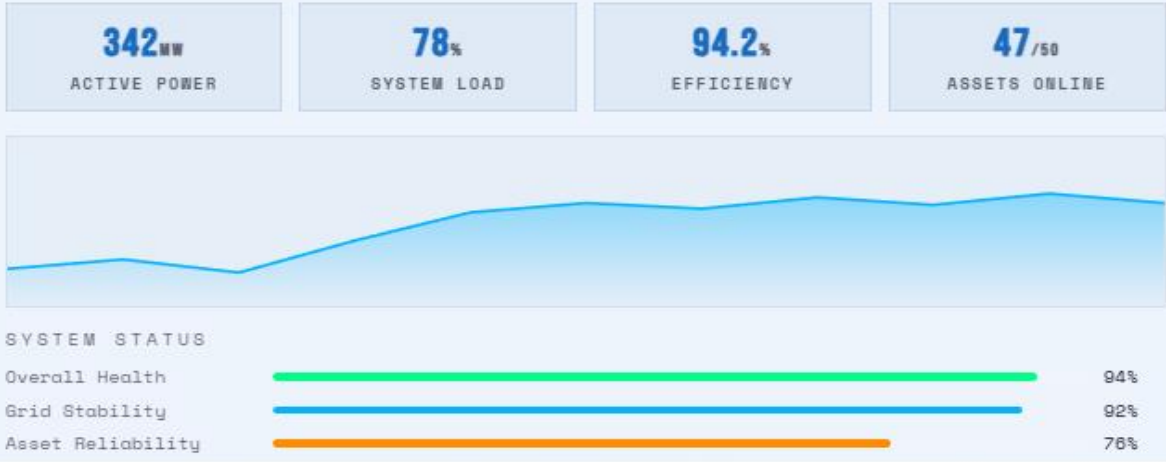
DEMO & OUTPUT

3D GRID SIMULATION

DIGITAL TWIN



• SYSTEM ONLINE 99.8%



EHV Substation Dashboard

Real-time analytics for 400/220 kV substation with live energy flow trends and asset health overview.

Real-Time Monitoring — SCADA & IoT Streams



Substation Schematic

Live data streams from SCADA systems and IoT sensors — voltage, current, temperature, frequency all tracked.

Failure & Recovery Simulations



Transformer Fault Scenario Output

Generation, Load, and Stability metrics during simulated T1 failure — showing AI-driven recovery at t=30s.

IMPACT.

- **Reduced Power Outages**

→ Early detection prevents failures before they occur

- **Lower Maintenance Costs**

→ Shift from reactive to predictive maintenance model

- **Faster Fault Response**

→ Automated actions reduce downtime significantly

- **Improved Grid Reliability**

→ Stable and efficient power distribution at scale

- **Renewable Energy Integration**

→ Handles dynamic and fluctuating supply from solar/wind

FUTURE.

- **Real IoT Sensor Integration**

→ Replace simulated data with live grid data feeds

- **Advanced AI Models — LSTM / Deep Learning**

→ Improve prediction accuracy over time with more data

- **Smart City Deployment**

→ Scalable architecture for large urban infrastructure

- **Cloud-Based Real-Time Systems**

→ Full deployment on AWS for enterprise scalability

- **Autonomous Energy Management**

→ Fully self-operating intelligent grid systems

THANK

YOU

PURAB GUPTA

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RUJUL BAKSHI

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SAHIL MALLICK

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AJAY KUMAR